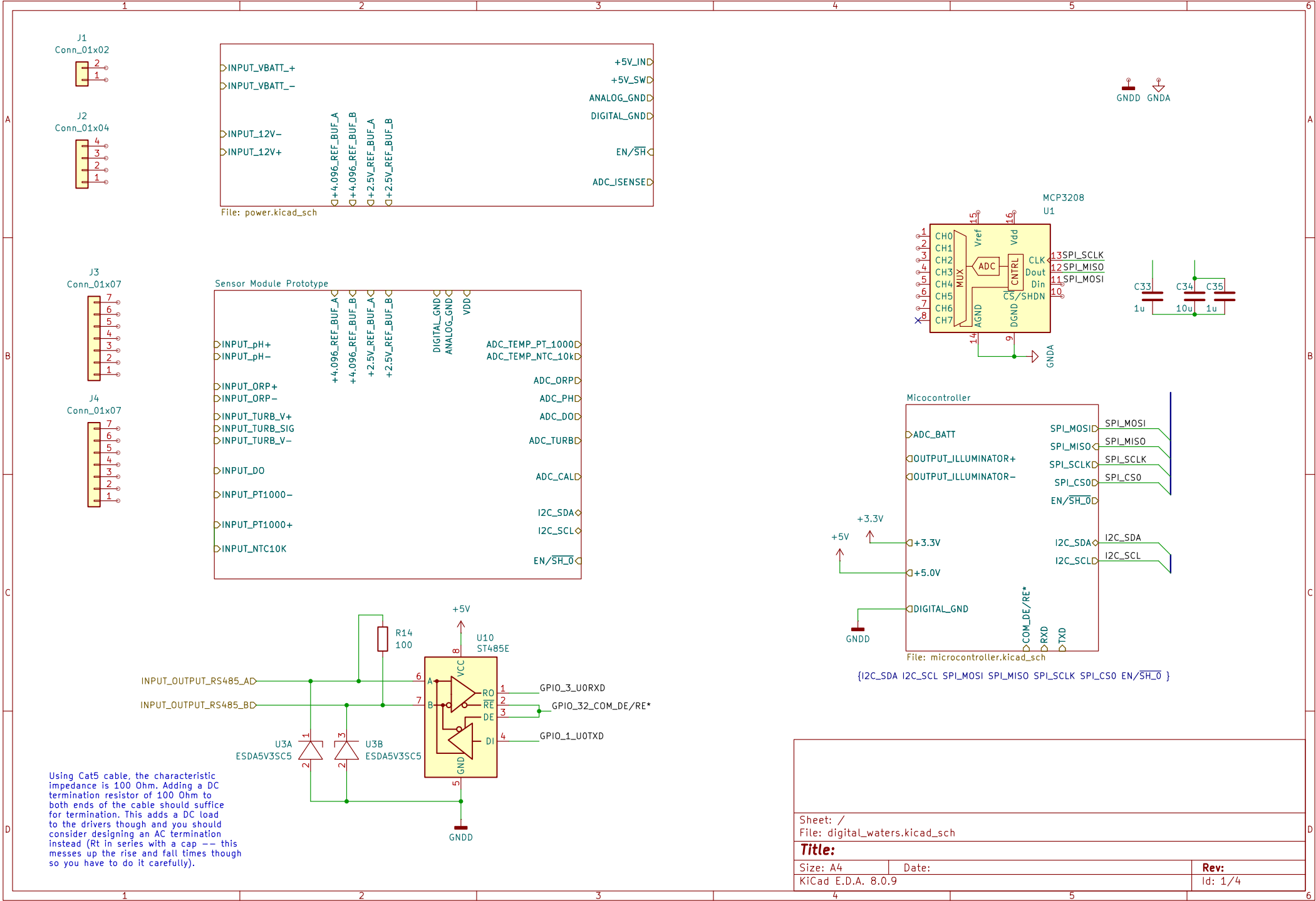
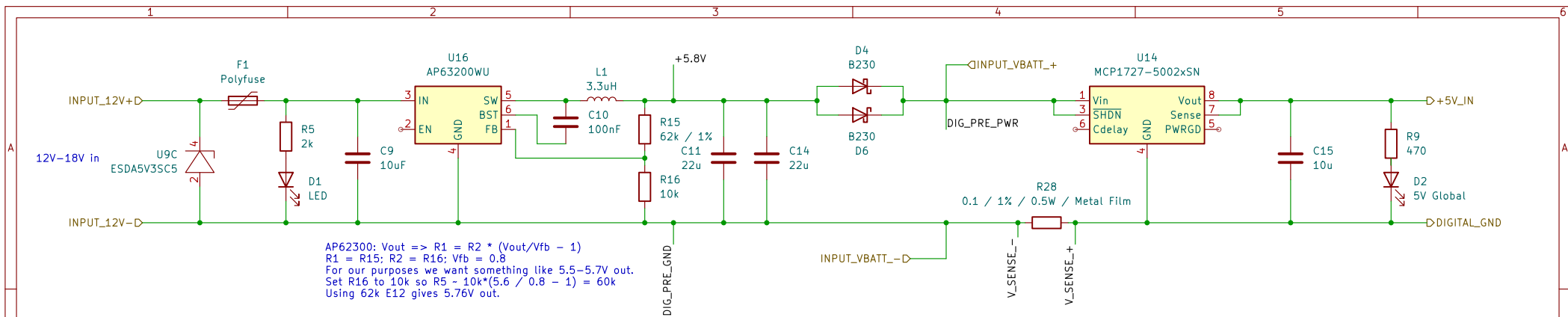
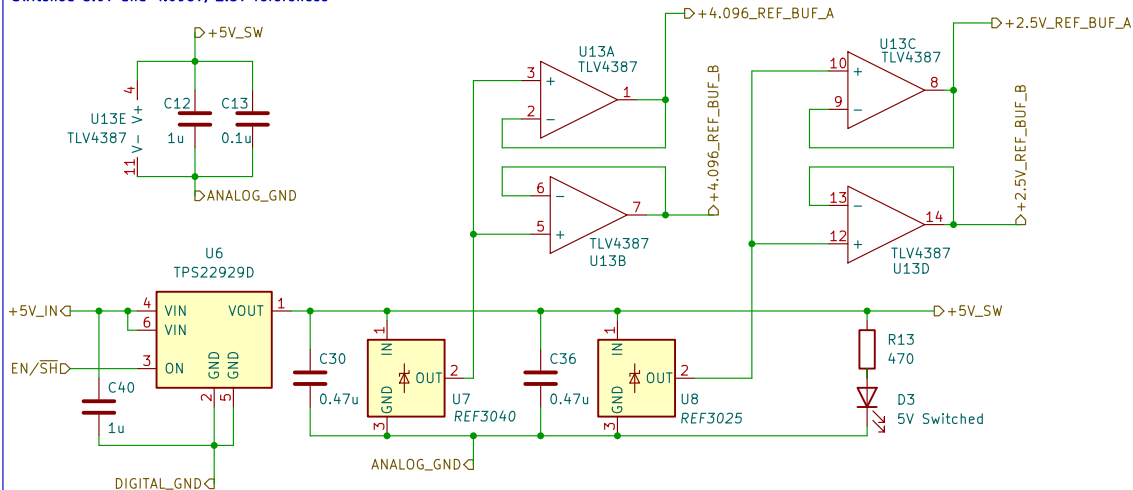


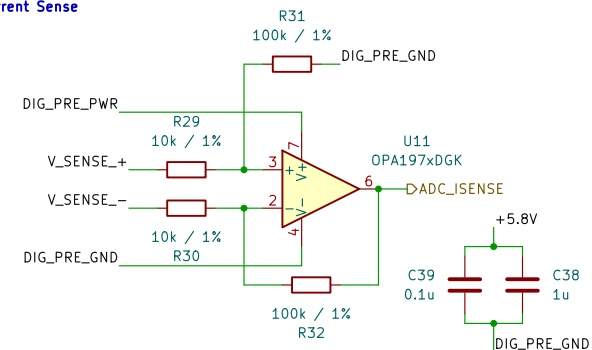




Switched 5.0V and 4.096V, 2.5V references



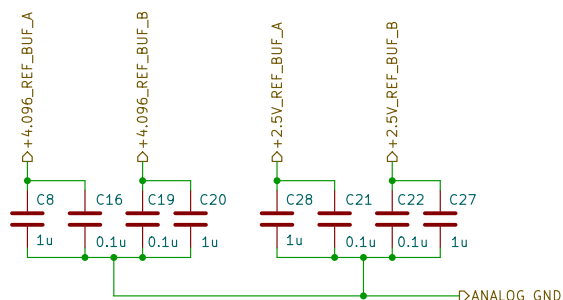
Current Sense



Max current is 1A, $R_{sense} = 0.1$
 so max power dissipation is 100mW
 through R_{sense} .
 @20mA $V_{drop, R_{sense}} = 2mV$
 @1A $V_{drop, R_{sense}} = 100mV$
 Gain = 10 for [20mV, 1V] output

TODO: U11 is referenced to
 DIG_PRE_GND so its output will
 be offset by the drop across
 R28 (R_{sense}). For example at
 $I = 20mA$ R_{sense} will have a drop of 2mV
 and U11 will output 20mV; however,
 this is referenced to DIG_PRE_GND which is
 2mV lower than DIGITAL_GND and so will
 be read as 18mV.
 Likewise with 1Amp, R_{sense} will drop 100mV,
 U11 will output 1V but referenced to
 DIG_PRE_GND which will be 100mV lower than
 DIGITAL_GND and so will be read as 900mV.

Reference Decoupling



AGND and DGND are tied together
 internally to the MCP320X, this
 connection should be as close to
 the chip as possible.



Sheet: /Power/
 File: power.kicad_sch

Title:

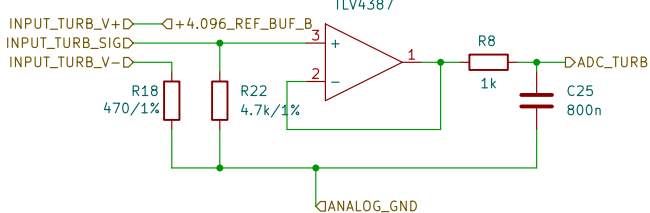
Size: A4
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Date:

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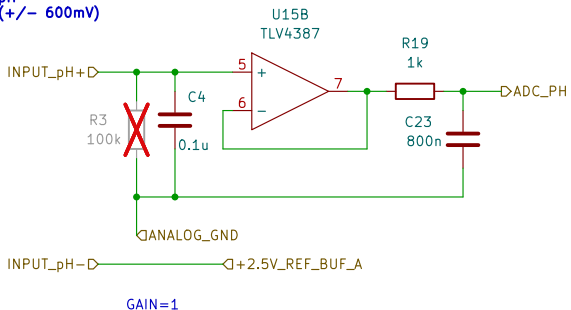
Id: 2/4

Turbidity

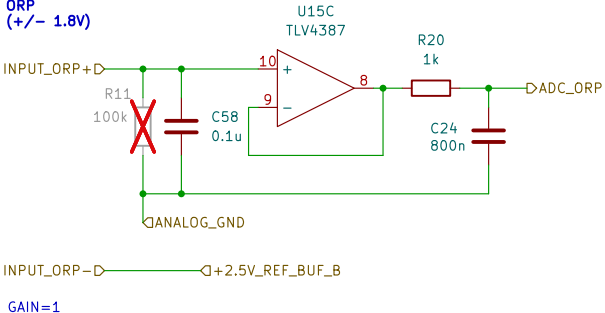


Resistor values are from the
datasheet.

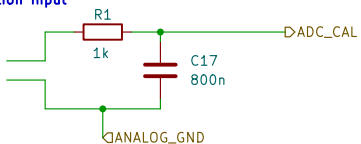
pH (+/- 600mV)



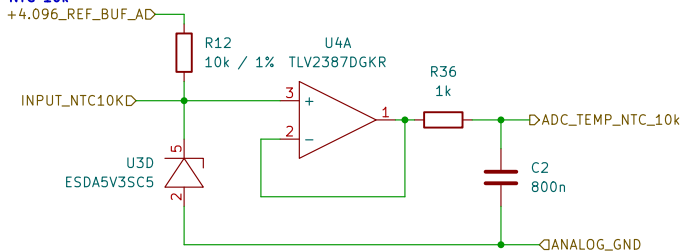
ORP (+/- 1.8V)



Calibration Input

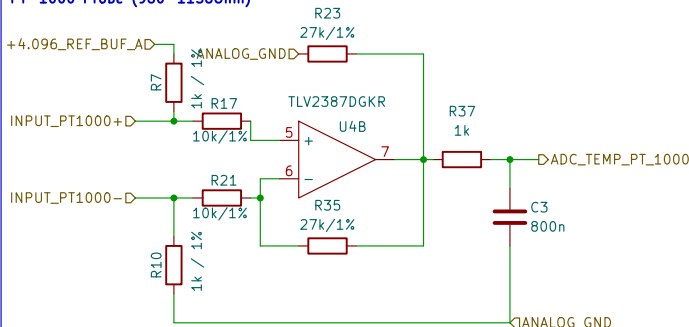


Ext Temperature Sense NTC 10k



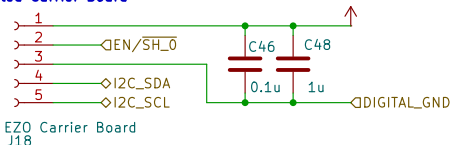
Assuming a B[25/50] of 4250 (from ERTJ thermistor manual)
@0 R=3.582*10k = 35820
@25 R=1 * 10k = 10k
@100 R=0.5353 * 10k = 5353
With a top resistor of 10k and a bias of 4.096 we have an
output range of [3.451, 1.4281] delta = 2.023V range.
This is about 0.020mV / C for that range. That corresponds
to about 20 LSB / C less nonlinearity.

Ext Temperature Sense PT-1000 Probe (960-1156Ohm)



Gain = 27
1k precision resistor gives
@0C R=1k
@25C R=1097.3
@40C R=1155.4
With a top resistor of 1k and a bias of 4.096 we have an
output range of [2.048, 2.196] delta = 0.148V range.
This is about mV / C which corresponds to about
LSB / C.

Atlas Scientific Isolated Carrier Board

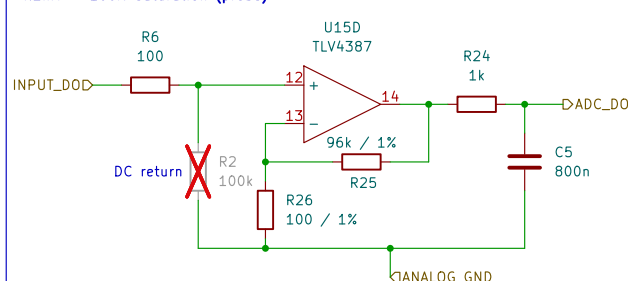


EZO Carrier Board
J18

NOTE!!! The header socket is flipped for mounting the
carrier board upsidedown.

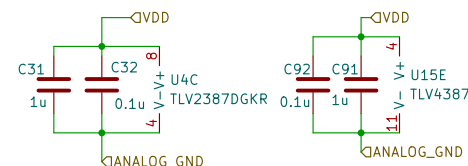
This is initially for the Atlas Scientific Conductivity board.
It is required to initialize the EC-EZO module to I2C comms.
See the EZO-EC datasheet for more information.

Dissolved Oxygen (Analog) 4.2mV = 100% saturation (probe)



Note that the Atlas Scientific DO board is simply a
5-3.3V regulator coupled with a non-inverting amp with
10x gain on it. 100% saturation is 42mV out of the
A.S. board so I'm assuming the probe outputs a max of
4.2mV. We will simply use a 960X gain stage here giving
us very nearly full scale at 100% saturation.

GAIN = 960x



Sheet: /Sensor Module Prototype/
File: sensor_module_4.kicad_sch

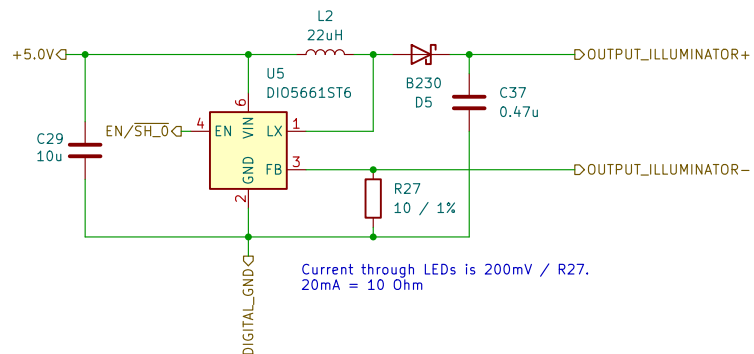
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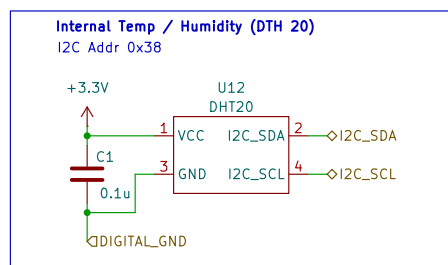
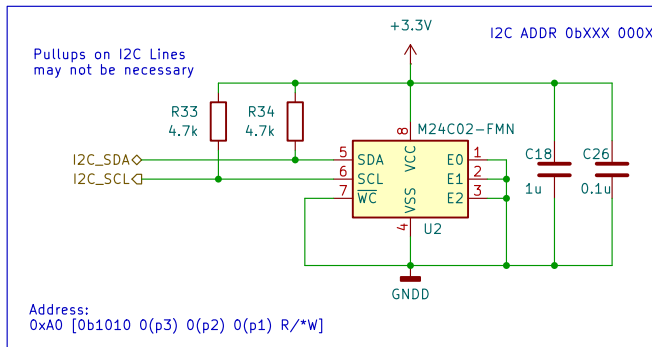
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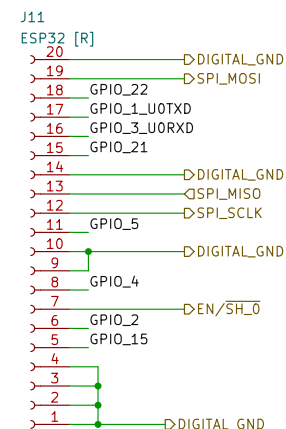
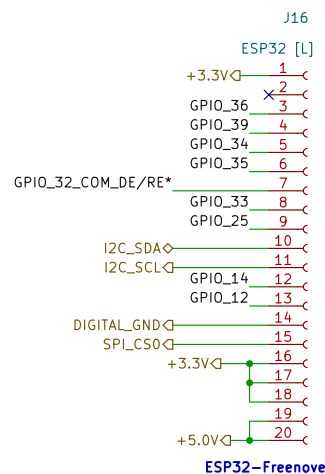
We need a powered output and switch for the LED ring. Ideally, a constant-current LED driver would be used although I'd like to keep the option open to use a multi-spectral led source which will probably want one channel per LED. For now, maybe just a constant-current serial output?



Current through LEDs is $200\text{mV} / R_{27}$.
 $20\text{mA} = 10\text{ Ohm}$



GPIO_32_COM_DE/RE* — COM_DE/RE*
 GPIO_1_U0TXD — TXD
 GPIO_3_U0RXD — RXD



NOTE: The ESP32 carrier board has headers on the bottom. Thus the carrier board header socket pinout is identical to the ESP32 board as viewed from above.

Sheet: /Microcontroller/
 File: microcontroller.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 8.0.9

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Id: 6/4